Figure 12b · One epoch, full data

RTX 6000 Ada · 32 test cases per family · Weight EMA

Native screens: shared timing proxy. Fast: directly evaluated and timed.

Stokes
T = 20
10⁻¹
10⁻²
10⁻³
10⁻⁴
0.0
0.8
1.6
2.4
M2
M3
Learning rate 1e−5 0.03444%
Control 0.02823%
Reflection augmentation 0.02817%
Multiplier skip 0.02762%
Momentum × ¼ 0.02739%
Momentum × ½ 0.02728%
Mode balancing × ⅓ 0.02682%
Longer LR plateau 0.02413%
Momentum ramp 0.01713%
Learning rate 4e−5 0.01684%
Adam β₂=.99 · seed 1 0.01642%
Log multiplier inputs 0.0162%
Parent · seed 1 0.01618%
Adam β₂=.99 · seed 1 · fast 0.01581%
Grad EMA .5 + β₁=.5 · seed 1 0.0154%
LR 6e−5 + momentum × ½ 0.01496%
Head learning rate × 2 0.01463%
Combined + ¼-epoch ramp 0.01461%
Data modes ≤ 128 0.01441%
Adam β₁=.7 0.01407%
LR 6e−5 + EMA .995 0.01388%
Adam β₁=.5 · fast 0.01364%
Gradient EMA .5 + Adam β₁=.5 0.01322%
LR 4e−5 + momentum × ¼ 0.01313%
LR 4e−5 + momentum × ½ 0.01306%
Mode-balance floor × 10 0.01281%
Gradient EMA .25 0.01277%
LR 8e−5 + Adam β₁=.5 0.0125%
Combined + EMA 0.995 0.012%
Adam β₂=.99 + LR 8e−5 · seed 1 · fast 0.01174%
Gradient EMA .5 + 20-step teacher 0.01088%
Gradient EMA .5 0.01061%
Gradient EMA .5 + head × 2 0.01018%
Datum augmentation · 2.5% RMS · fast 0.0101%
Gradient EMA .5 + LR 8e−5 0.00936%
Adam β₂=.99 0.008716%
Adam β₂=.99 + MB floor × 10 0.008685%
Adam β₂=.99 · fast 0.008601%
Adam β₂=.99 + MB floor × 10 · fast 0.008455%
Whitened surface inputs 0.007874%
Depth-conditioned surface head · fast 0.007802%
Adam β₂=.99 + LR 1e−4 · fast 0.007545%
LR 8e−5 + momentum × 2 · fast 0.007272%
Batch 64 · Hadamard × 4 · seed 1 · fast 0.006992%
Adam β₂=.99 + LR 8e−5 · fast 0.006716%
Gradient EMA .5 · seed 1 0.006656%
Batch 64 · softened curvature · fast 0.006567%
Amplitude targets · full band 0.00613%
Amplitude targets · ≤128 · fast 0.006075%
Batch 64 · Hadamard flow probes · fast 0.005965%
Amplitude targets · full band · fast 0.005926%
Amplitude targets · ≤128 0.005903%
LR 8e−5 + batch 256 · fast 0.005738%
Batch 128 · seed 1 · fast 0.005691%
Batch 128 · seed 1 · exact G₀ 0.005567%
Batch 512 · seed 1 · fast 0.00549%
Batch 256 · seed 1 · fast 0.005469%
Batch 64 · seed 1 · fast 0.005185%
LR 8e−5 + batch 128 · fast 0.005029%
Context width 16 · fast 0.004848%
LR 8e−5 + batch 512 · fast 0.004746%
Batch 64 · Hadamard × 16 · fast 0.004557%
Batch 32 · Hadamard × 4 · seed 1 · fast 0.004106%
LR 8e−5 + batch 64 · fast 0.004073%
Batch 64 · Hadamard × 4 · fast 0.00399%
Batch 32 · Hadamard × 4 · fast 0.003842%
Batch 32 · HD × 4 · xi norm · seed 1 · fast 0.002517%
Batch 32 · HD × 4 · modes ≤128 · fast 0.001864%
Batch 32 · HD × 4 · xi norm · grid512 0.001737%
Batch 32 · HD × 4 · xi norm · fast 0.001736%
Batch 32 · HD × 4 · ≤128 · seed 1 · fast 0.001525%
Tanaka
T = 200
10⁰
10⁻¹
10⁻²
0
8
16
24
M2
M3
Mode balancing × ⅓ 1.182%
Learning rate 1e−5 0.953%
Reflection augmentation 0.935%
Control 0.7349%
Multiplier skip 0.723%
Momentum × ¼ 0.7196%
Learning rate 4e−5 0.6694%
Momentum × ½ 0.6673%
LR 4e−5 + momentum × ¼ 0.5083%
Head learning rate × 2 0.4701%
Log multiplier inputs 0.4334%
Longer LR plateau 0.4024%
LR 6e−5 + momentum × ½ 0.395%
Combined + ¼-epoch ramp 0.3855%
LR 4e−5 + momentum × ½ 0.3827%
Data modes ≤ 128 0.3753%
LR 6e−5 + EMA .995 0.3563%
Combined + EMA 0.995 0.3479%
Momentum ramp 0.3433%
Gradient EMA .5 · seed 1 0.3354%
Whitened surface inputs 0.3288%
Grad EMA .5 + β₁=.5 · seed 1 0.2946%
Parent · seed 1 0.2942%
Gradient EMA .5 + head × 2 0.2834%
Gradient EMA .5 + 20-step teacher 0.2494%
Context width 16 · fast 0.2433%
Adam β₂=.99 + MB floor × 10 · fast 0.2387%
Adam β₂=.99 + MB floor × 10 0.2384%
Gradient EMA .5 0.235%
Gradient EMA .5 + LR 8e−5 0.2317%
LR 8e−5 + momentum × 2 · fast 0.221%
LR 8e−5 + Adam β₁=.5 0.2138%
Adam β₂=.99 + LR 1e−4 · fast 0.199%
Gradient EMA .5 + Adam β₁=.5 0.1933%
Adam β₁=.5 · fast 0.1932%
Adam β₂=.99 · seed 1 · fast 0.1928%
Adam β₂=.99 · seed 1 0.1925%
Amplitude targets · ≤128 0.19%
Amplitude targets · ≤128 · fast 0.1899%
Gradient EMA .25 0.182%
Adam β₂=.99 · fast 0.1799%
Adam β₂=.99 0.1789%
Amplitude targets · full band · fast 0.1757%
Amplitude targets · full band 0.1755%
Adam β₁=.7 0.1752%
Depth-conditioned surface head · fast 0.174%
Batch 512 · seed 1 · fast 0.1722%
Adam β₂=.99 + LR 8e−5 · fast 0.1669%
Datum augmentation · 2.5% RMS · fast 0.1644%
Mode-balance floor × 10 0.1628%
LR 8e−5 + batch 512 · fast 0.1512%
Adam β₂=.99 + LR 8e−5 · seed 1 · fast 0.1387%
LR 8e−5 + batch 128 · fast 0.1305%
LR 8e−5 + batch 256 · fast 0.1106%
Batch 256 · seed 1 · fast 0.1105%
Batch 128 · seed 1 · fast 0.08213%
Batch 128 · seed 1 · exact G₀ 0.08137%
LR 8e−5 + batch 64 · fast 0.07513%
Batch 64 · Hadamard × 16 · fast 0.07379%
Batch 64 · softened curvature · fast 0.07321%
Batch 64 · Hadamard flow probes · fast 0.07099%
Batch 64 · Hadamard × 4 · fast 0.06887%
Batch 64 · seed 1 · fast 0.06556%
Batch 64 · Hadamard × 4 · seed 1 · fast 0.06086%
Batch 32 · HD × 4 · xi norm · seed 1 · fast 0.05908%
Batch 32 · Hadamard × 4 · seed 1 · fast 0.05374%
Batch 32 · Hadamard × 4 · fast 0.04868%
Batch 32 · HD × 4 · xi norm · fast 0.03882%
Batch 32 · HD × 4 · xi norm · grid512 0.03881%
Batch 32 · HD × 4 · ≤128 · seed 1 · fast 0.03578%
Batch 32 · HD × 4 · modes ≤128 · fast 0.02655%
Median final surface error (%)
Benjamin–Feir
T = 200
10⁰
10⁻¹
0
8
16
24
M2
M3
Learning rate 1e−5 1.753%
Reflection augmentation 1.499%
Control 1.447%
Mode balancing × ⅓ 1.412%
Multiplier skip 1.349%
Learning rate 4e−5 1.32%
Momentum × ¼ 1.297%
Momentum × ½ 1.279%
LR 4e−5 + momentum × ¼ 1.216%
Combined + EMA 0.995 1.146%
Combined + ¼-epoch ramp 1.139%
LR 4e−5 + momentum × ½ 1.135%
LR 6e−5 + EMA .995 1.058%
Momentum ramp 1.049%
LR 6e−5 + momentum × ½ 1.006%
Longer LR plateau 0.9989%
Data modes ≤ 128 0.9985%
Log multiplier inputs 0.9948%
Head learning rate × 2 0.9004%
Parent · seed 1 0.8859%
Gradient EMA .5 + head × 2 0.7403%
Gradient EMA .5 0.6623%
Gradient EMA .5 · seed 1 0.6533%
Gradient EMA .5 + 20-step teacher 0.6503%
Gradient EMA .5 + LR 8e−5 0.5803%
Adam β₁=.7 0.5669%
Gradient EMA .25 0.5559%
Adam β₁=.5 · fast 0.5222%
Gradient EMA .5 + Adam β₁=.5 0.5209%
Whitened surface inputs 0.5202%
Mode-balance floor × 10 0.5099%
Datum augmentation · 2.5% RMS · fast 0.4926%
Grad EMA .5 + β₁=.5 · seed 1 0.4854%
Context width 16 · fast 0.48%
Amplitude targets · full band · fast 0.4722%
Amplitude targets · full band 0.4719%
Amplitude targets · ≤128 0.4685%
Amplitude targets · ≤128 · fast 0.4663%
LR 8e−5 + Adam β₁=.5 0.4569%
Depth-conditioned surface head · fast 0.4545%
Adam β₂=.99 + LR 8e−5 · seed 1 · fast 0.4514%
Adam β₂=.99 + MB floor × 10 0.4485%
Adam β₂=.99 + MB floor × 10 · fast 0.4479%
Adam β₂=.99 0.4436%
Adam β₂=.99 · fast 0.4432%
LR 8e−5 + momentum × 2 · fast 0.4301%
Adam β₂=.99 · seed 1 0.4174%
Adam β₂=.99 · seed 1 · fast 0.4169%
Adam β₂=.99 + LR 1e−4 · fast 0.4022%
Adam β₂=.99 + LR 8e−5 · fast 0.392%
LR 8e−5 + batch 512 · fast 0.3622%
Batch 512 · seed 1 · fast 0.3157%
LR 8e−5 + batch 256 · fast 0.2977%
Batch 256 · seed 1 · fast 0.2605%
Batch 128 · seed 1 · exact G₀ 0.2178%
Batch 128 · seed 1 · fast 0.2177%
Batch 64 · Hadamard flow probes · fast 0.2129%
Batch 32 · HD × 4 · xi norm · fast 0.1875%
Batch 32 · HD × 4 · xi norm · grid512 0.1875%
LR 8e−5 + batch 128 · fast 0.1673%
Batch 64 · softened curvature · fast 0.1598%
Batch 64 · Hadamard × 16 · fast 0.1595%
Batch 64 · Hadamard × 4 · fast 0.1593%
Batch 64 · seed 1 · fast 0.1592%
LR 8e−5 + batch 64 · fast 0.1555%
Batch 32 · HD × 4 · xi norm · seed 1 · fast 0.1542%
Batch 32 · HD × 4 · modes ≤128 · fast 0.154%
Batch 32 · HD × 4 · ≤128 · seed 1 · fast 0.1467%
Batch 64 · Hadamard × 4 · seed 1 · fast 0.1421%
Batch 32 · Hadamard × 4 · fast 0.1335%
Batch 32 · Hadamard × 4 · seed 1 · fast 0.1223%
JONSWAP / TMA
T = 20
10⁰
10⁻¹
10⁻²
0.0
0.8
1.6
2.4
M2
M3
Learning rate 1e−5 0.8097%
Mode balancing × ⅓ 0.7267%
Momentum × ¼ 0.7177%
Control 0.7175%
Reflection augmentation 0.7168%
Multiplier skip 0.7054%
Momentum × ½ 0.6803%
Combined + ¼-epoch ramp 0.6093%
LR 4e−5 + momentum × ½ 0.608%
Combined + EMA 0.995 0.5905%
Log multiplier inputs 0.5885%
LR 4e−5 + momentum × ¼ 0.5512%
Learning rate 4e−5 0.5039%
LR 6e−5 + EMA .995 0.4445%
LR 6e−5 + momentum × ½ 0.4444%
Head learning rate × 2 0.4308%
Longer LR plateau 0.4293%
Gradient EMA .5 · seed 1 0.4283%
Momentum ramp 0.427%
Parent · seed 1 0.4179%
Data modes ≤ 128 0.4143%
Grad EMA .5 + β₁=.5 · seed 1 0.3867%
Gradient EMA .25 0.3436%
Adam β₂=.99 · seed 1 · fast 0.3189%
Adam β₂=.99 · seed 1 0.3189%
Adam β₁=.7 0.3079%
Gradient EMA .5 + Adam β₁=.5 0.3051%
Adam β₁=.5 · fast 0.3051%
Mode-balance floor × 10 0.3049%
Whitened surface inputs 0.2801%
Gradient EMA .5 0.2746%
Gradient EMA .5 + 20-step teacher 0.2745%
Adam β₂=.99 + LR 8e−5 · seed 1 · fast 0.2742%
LR 8e−5 + Adam β₁=.5 0.2737%
Gradient EMA .5 + LR 8e−5 0.2721%
Context width 16 · fast 0.2705%
Amplitude targets · full band 0.2701%
Amplitude targets · full band · fast 0.2701%
Datum augmentation · 2.5% RMS · fast 0.2592%
Gradient EMA .5 + head × 2 0.2585%
Amplitude targets · ≤128 0.2562%
Amplitude targets · ≤128 · fast 0.2561%
Depth-conditioned surface head · fast 0.2554%
Adam β₂=.99 0.2526%
Adam β₂=.99 · fast 0.2526%
Adam β₂=.99 + LR 8e−5 · fast 0.2506%
Adam β₂=.99 + MB floor × 10 0.2486%
Adam β₂=.99 + MB floor × 10 · fast 0.2485%
LR 8e−5 + momentum × 2 · fast 0.2472%
Adam β₂=.99 + LR 1e−4 · fast 0.2421%
Batch 512 · seed 1 · fast 0.2112%
LR 8e−5 + batch 512 · fast 0.1766%
LR 8e−5 + batch 256 · fast 0.1662%
Batch 256 · seed 1 · fast 0.1532%
LR 8e−5 + batch 128 · fast 0.1434%
Batch 64 · Hadamard flow probes · fast 0.1434%
Batch 64 · Hadamard × 4 · seed 1 · fast 0.1254%
Batch 32 · Hadamard × 4 · seed 1 · fast 0.1239%
Batch 64 · seed 1 · fast 0.1192%
Batch 64 · Hadamard × 4 · fast 0.1189%
Batch 128 · seed 1 · exact G₀ 0.1178%
Batch 128 · seed 1 · fast 0.1178%
LR 8e−5 + batch 64 · fast 0.1173%
Batch 64 · softened curvature · fast 0.1156%
Batch 64 · Hadamard × 16 · fast 0.1035%
Batch 32 · HD × 4 · xi norm · seed 1 · fast 0.1014%
Batch 32 · HD × 4 · xi norm · grid512 0.09909%
Batch 32 · HD × 4 · xi norm · fast 0.09909%
Batch 32 · Hadamard × 4 · fast 0.08944%
Batch 32 · HD × 4 · ≤128 · seed 1 · fast 0.06454%
Batch 32 · HD × 4 · modes ≤128 · fast 0.06239%
Single-rollout runtime (seconds)